

Pandas Cheatsheet — 40 Essential Functions

Data Loading & Inspection	
pd.read_csv("file.csv")	Load CSV
pd.read_json("data.json")	Load JSON
df.info()	Schema + nulls
df.head(10)	Preview rows
df.sample(5)	Random sample
df.shape	Rows/columns
df.describe()	Summary stats

Cleaning & Transforming	
df.dropna(subset=['id'])	Drop missing
df.fillna(0)	Fill missing
df.drop_duplicates(subset=['id'])	Deduplicate
df.duplicated()	Check duplicates
df.columns.difference(['id'])	Exclude columns
df.reset_index(drop=True)	Reset index
df.rename(columns={'a':'A'})	Rename
df.drop(columns=['col'])	Drop column
df.replace({'A':1})	Replace values
np.where(df['x']>0,1,0)	Conditional column

Aggregation & Grouping	
df.groupby('cat')['sales'].sum()	Group + sum
df.groupby('id').agg({'a':'sum', 'b':'count'})	Multi-agg
df['id'].count()	Count
df['user_id'].nunique()	Unique count
df['country'].value_counts()	Frequency
df.sort_values('sales', ascending=False)	Sort

Selecting & Filtering	
df['col']	Select column
df[['a', 'b']]	Select multiple columns
df[df['age'] > 30]	Boolean filter
df[(df['age']>18) & (df['country']=='US')]	Multi-condition
df.query("country=='US'")	SQL-style filter
df[df['age'].between(20,30)]	Range filter

Type Handling	
df['id'].astype('Int64')	Cast type
pd.to_datetime(df['ts'])	To datetime
df['ts'].dt.date	Extract date
df['ts'].dt.year	Extract year

Joining & Combining	
pd.merge(a, b, on='id', how='left')	SQL join
df1.join(df2, how='left')	Index join
pd.concat([df1, df2])	Stack DataFrames

Advanced Operations	
df['ma7'] = df['sales'].rolling(7).mean()	Rolling window
df['r'] = df.groupby('cat')['sales'].rank()	Rank
df.pivot_table(values='sales', index='cat', columns='region')	Pivot
df['x2'] = df['x'].apply(lambda x: x*2)	Apply
df['email'].str.lower()	String ops
df.loc[df['age']<0, 'age'] = 0	Update rows